Rusumba Bukanga

AI Product Engineer | Full-Stack (TypeScript, Python)

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SUMMARY

AI product engineer with 3+ years shipping production software, most recently LLM features on Claude: an AI grading platform with RAG, tiered OCR and cost telemetry, and a streaming AI tutor. Cofounder and CTO of Cleverly AI. At Alpha Group I built the company SSO gateway and insurance systems that cut policy processing time by 60% and data-entry errors by 85%. Strongest in TypeScript, Node, Python and Postgres on AWS.

SKILLS

- **AI:** Claude API, prompt engineering and caching, RAG and embeddings, OCR pipelines, provider-agnostic LLM layers, cost telemetry
- **Languages:** TypeScript, JavaScript, Python, SQL
- **Frameworks:** Next.js, React, Node.js, Express, FastAPI
- **Data:** PostgreSQL, Supabase, DynamoDB, MongoDB, BigQuery
- **Cloud and delivery:** AWS (Lambda, API Gateway, Cognito, EventBridge, S3, Bedrock), GCP, Docker, CI/CD, Vercel, Render
- **Auth and security:** OAuth2/PKCE, JWT, row-level security, access control, dependency audits
- **Testing:** Jest, integration testing

EXPERIENCE

Cofounder and CTO, Cleverly AI | Oct 2025 to Present

- Built an AI-assisted grading platform for South African schools, colleges and universities (Next.js, Express, Supabase): rubric-aligned, evidence-linked grading with teacher review before results are released. Currently in pilot with five schools.
- Designed a vendor-neutral AI provider layer running Claude, with RAG and local embeddings, so models can be swapped without changing product code.
- Built a tiered OCR pipeline that tries free text extraction first and uses vision-based handwriting recognition only when needed.
- Added prompt caching, model tiering and cross-provider cost tracking to keep unit economics sustainable as usage grows.
- Ran a security hardening pass covering access control, rate limiting and POPIA-aligned data handling.
- Own CI, automated testing and deployment on Vercel and Render. Manage hiring, budget and vendors. Appointed Director.

CTO, 123Tutors | Aug 2025 to Present

- Led the migration from Bubble to Next.js and TypeScript on Prisma, Supabase, Vercel and AWS (S3, Lambda, EventBridge). The platform has 9 years of trading history and 2,300+ active tutors.
- Shipped an AI tutor on Claude via AWS Bedrock: streaming chat, intent classification, worked examples and generated mindmaps and diagrams.
- Reviewed the codebase and security posture and fixed the critical production risks.
- Own CI/CD, row-level security authorization and integrations with Paystack, Twilio and the WhatsApp Cloud API.
- Directed the roadmap for the tutor-matching engine and institutional reporting.

Junior Software Engineer and ML Engineer, 123Tutors | Apr 2023 to Sep 2024

- Built cloud-native services in Ruby on Rails and FastAPI, with CI/CD and containerised deployments on Google Cloud.
- Built REST APIs and automated deployment pipelines for machine learning models.

Full Stack Developer, Alpha Group | Aug 2025 to Present

Alpha Group runs Alpha Insure, The 11th Floor restaurant and Body Action Gym.

- Designed and built the central SSO authentication gateway used across the group's applications (AWS Lambda, API Gateway, Cognito, DynamoDB, KMS-signed JWTs).
- Alpha Insure: built CRM, quote calculator, policy management and payment workflows, plus APIs to underwriting partners, banks and reinsurers. Cut policy processing time by 60% and manual data-entry errors by 85%.
- Lead developer of an internal forms platform that converts legacy Adobe XFA insurance forms into web forms, with a form builder, PDF import, e-signing through SigniFlow and AI-assisted agreement import on Amazon Bedrock. Added a CI quality gate so form conversions are regression-tested before release.

- Built a UAT and regression runner for the Akur8 pricing engine, documented so another engineer can rebuild it.
- Authored an audit of 100+ repositories documenting architecture, dependencies and operational risk for business continuity.
- Led the broker onboarding portal and built an HR portal integrated with Microsoft Graph.
- Built the automated recurring debit order run on AWS (Lambda, EventBridge, S3, SES/SNS) with bank reconciliation.
- Built the booking system for The 11th Floor and the membership and events platform for Body Action Gym.

NOC Support Engineer, Reflex Solutions (contract) | Dec 2024 to Jul 2025

- Ran 24/7 network operations for enterprise clients on FortiGate, F5 and Cisco, and automated diagnostics with Python.

SELECTED PROJECTS

- **Cleverly AI Grader:** evidence-linked grading on Claude with RAG, tiered OCR and cost telemetry.
- **Central SSO gateway:** OAuth2/PKCE, Cognito and KMS-signed JWTs serving a whole application ecosystem.
- **123Tutors migration and AI tutor:** Bubble to Next.js/TypeScript, with a streaming Claude tutor. Live: 123tutors.co.za
- **Forms and e-signing platform:** legacy Adobe XFA forms rebuilt as web forms with a builder, PDF import and SigniFlow e-signing, using Amazon Bedrock for AI-assisted agreement import.

EDUCATION AND CERTIFICATIONS

- BCom, Logistics Management, University of Johannesburg
- SLP, Quantitative Finance, University of Johannesburg
- HCIA, CCNA, ITIL, cloud fundamentals (AWS, GCP, Azure)

LANGUAGES

English (fluent), French (fluent)